MIL-D-38636A(USAF)
23 September 1974

SUPERSEDING
MIL-D-38636(USAF)
25 May 1967

MILITARY SPECIFICATION

DISPENSER, BOMB AND ROCKET, SUU-20B/A, TRAINING TYPE AND GUN ASSEMBLY, BOMB EJECTOR, SUU-20B/A

This specification is mandatory for use by USAF

1. SCOPE

1.1 Scope. This specification establishes the requirements for fabrication and acceptance of the Dispenser, Bomb and Rocket, SUU-20B/A, hereafter referred to as the dispenser, and of the Gun Assembly, Bomb Ejector, hereafter referred to as the ejector, when procured separately from the dispenser.

2. APPLICABLE DOCUMENTS

2.1 Government Documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-A-8591	Airborne Stores and Associated Suspension Equipment, General Design Criteria For
MIL-Q-9858	Quality Control System Requirements
MIL-C-45662	Calibration System Requirements

STANDARDS

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-785	Requirements for Reliability Program (For Systems and Equipment)
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment

DRAWINGS

Air Force

66D13250-7	Simulator, SMU-75/A

FSC 1095

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66D13258-7 Harness, Wiring
66J13259-7 Installation, Simulator (Kit)
66D13260-1 Simulator, SMU-64/A
69D6070 Dispenser, Bomb and Rocket, SUU-20B/A
69E6100 Gun Assembly, Bomb Ejector

OTHER

Technical Manuals

TPO 1095-111-4657 Transportation Packaging Order, Dispenser, SUU-20B/A (Copies may be obtained from Warner Robins ALC/DSPCB, Robins AFB GA 31098)
T. O. 11B29-3-28-1 Operations, Field Maintenance and Overhaul Instructions With Illustrated Parts Breakdown for SUU-20 Bomb and Rocket Dispenser.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item Definition. The dispenser is a pod which can be carried on pylon/bomb racks of all fighter-bomber aircraft equipped with standard 14 inch or 30 inch suspension shackles. The pod has six (6) ejectors for force ejecting bombs which weigh from three (3) to twenty-five (25) pounds, and has four (4) rocket tubes for launching 2.75 inch FFAR rockets. Rotary selector switches in the electrical system control the firing mode of the bombs and rockets.

3.2 Preproduction Qualification. The dispensers and ejectors furnished under this specification shall be products that have been tested and passed the preproduction tests specified herein.

3.3 Components.

3.3.1 The SUU-20B/A dispenser shall comprise an integral assembly consisting of:

a. Center section assembly with 30 inch or 14 inch suspension lugs and provisions for electrical interface with all fighter-bomber aircraft.

b. Four (4) rocket tubes with detents for safely retaining rockets during aircraft takeoff, flight and landing operations.

c. Six (6) cartridge actuated bomb ejector assemblies equipped with adjustable "ice tong" type retaining arms for 3 inch, 3.87 inch or 4 inch diameter stores and with forward and aft adjustable sway braces.

d. Forward and aft cast aluminum end caps.

e. Electrical assembly consisting of wire harness, ballast resistors, mode selector switches for bombs and rockets and a firing circuit safety switch.

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3.4 Design and Construction. The dispenser and ejector shall be constructed and assembled in accordance with the requirements of Air Force Drawing 69D6070 and 69E6100, shall meet the design requirements of MIL-A-8591, and shall meet the test requirements of Section 4 of this specification.

3.4.1 Selection of Materials. Specifications and standards of all materials and parts, and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143.

3.4.2 Interchangeability. Dispenser and ejector parts which are functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable, the criteria listed in MIL-D-1000 shall apply and new drawings shall be prepared as required.

3.4.3 Workmanship. The dispenser and ejector shall be fabricated and finished in a manner which will insure that the criteria of appearance, fit, smoothness of operation, and adherence to specified tolerances are observed. These criteria must apply to all accessories and to all external nonfunctioning parts, such as name plates, as well as functional parts. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, machine finishes, and freedom of parts from burrs and sharp edges.

3.5 Performance. With the rocket mode selector switch energized, the dispenser shall be capable of firing four (4) 2.75 inch FFAR rockets in either single or ripple modes. With the bomb mode selector switch energized, the dispenser shall be capable of force ejecting six practice bombs (3 to 25 lbs) in either single, ripple or salvo modes. The dispenser shall be capable of flight speeds of up to mach 1.5 and shall be capable of withstanding constant or repeated structural loading up to 7.5 gravitational units (g) in any direction.

3.5.1 Reliability. The contractor shall initiate and maintain a reliability program in accordance with MIL-STD-785.

3.5.1.1 Reliability Levels. The following reliability levels shall be achieved and demonstrated by the contractor:

a. The demonstrated reliability level shall be 98.0 per cent and the demonstrated confidence level shall be 90 per cent for the complete dispenser assembly for 150 rocket firings and 250 practice bomb firings with no uncorrected failures.

b. Inadvertent Release - With proper adjustment of bomb ejector sway braces and rocket tube detents, there shall be no inadvertent release of bombs or rockets during the design service life of the dispenser (250 firing missions under service conditions).

c. Release Performance Requirements - With proper maintenance and replacement of parts, the achieved rate of failure shall be no failures in 250 bomb releases and 150 rocket firings under service conditions, excluding cartridge and rocket failures.

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3.5.2 Design Service Life. The dispenser shall function satisfactorily for 250 cartridge actuated bomb ejections and 150 rocket firings.

3.2.2.1 Bomb Ejection Velocity. The ejector shall demonstrate compliance with design and manufacturing requirements by successfully meeting the ejection velocity requirements of Section 4.

3.5.3 Special Weapon Provisions. When required by the procuring activity, a SMU-75/A Simulator, part number 66D13250-7, a SMU-64/A Simulator, part number 66D13260-1, and a wiring harness, part number 66D13258-7, (all furnished as GFE) shall be installed in the dispenser in conformance to kit drawing 66J13259-7.

3.5.4 Locked Shut Firing. The bomb ejector shall be capable of withstanding the stresses incurred when fired with a safety pin installed. The bomb shall not release and there shall be no damage to the ejector assembly. Damage is acceptable on the safety pin.

3.5.5 Electrical Circuit Safety Provisions. The dispenser bomb and rocket firing circuits shall be equipped with a mechanically actuated switch which when actuated shall interrupt the firing circuits and preclude any possibility of inadvertent release or firing of bombs or rockets during ground service operations.

3.6 Environmental Requirements. The dispenser shall be capable of proper operation without malfunction during or after, as applicable, the following environmental conditions:

a. High Temperature (160°F).

b. Low Temperature (-65°F).

c. Temperature Shock (160°F to -65°F).

d. Humidity up to 95 per cent, relative.

e. Icing conditions incident to service use.

f. Vibration incident to flight, transportation and handling.

g. Shock incident to flight, transportation and handling.

3.7 Preproduction Sample.

3.7.1 Dispenser Assembly. When specified by the contract, the contractor shall fabricate two (2) complete dispensers in accordance with Air Force drawing 69D6070 and subsidiary drawings for the purpose of undergoing the applicable preproduction and acceptance testing requirements of Section 4. The test samples shall have been produced with contractor manufacturing equipment, procedures and/or subcontractor parts which are to be used in contract production.

3.7.2 Ejector Unit Assemblies. (When procured separately from the dispenser.) When specified by the contract, the contractor shall fabricate four (4) complete ejector assemblies in accordance with Air Force drawing 69E6100 and

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subsidiary drawings for the purpose of undergoing the applicable preproduction and acceptance testing requirements of Section 4. The test samples shall have been produced with contractor manufacturing equipment, procedures and/or subcontractor parts which are to be used in contract production.

3.8 Identification of Product. The dispenser/ejector and all component parts shall be marked for identification in accordance with requirements of applicable drawings and MIL-STD-130.

3.9 Government Furnished Property. When specified in the contract, the Government will furnish the following items to the contractor for preproduction and acceptance test purposes.

Item Number	Part Number	Description	Quantity
1	Air Force Part Number 69F30301	BDU-33 Practice Bomb (Inert)	6
2	Air Force Part Number 60D90955-5	MK 106 Practice Bomb (Inert)	12
3	Part Number J30310110, Mfr Code 100001	MK 43, 2.75 FFAR Rocket Motor (Inert)	4
4	Part Number 1340H835, Mfr Code 99999	MK 61 Warhead (Inert)	4
5	Army Part Number 9242550 or 9242551	MK 40, 2.75 FFAR Rocket Motor (Live) With Warhead (Inert) Shipped Four per Box	11 Boxes
6	Part Number 10-3400, Mfr Code 80236	ARD 863-1 Cartridge	As Stated in Contract

4. QUALITY ASSURANCE

4.1 Responsibility for Inspection and Testing. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection and testing requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the procuring activity. The Government reserves the right to perform or witness any of the inspections or tests set forth in this specification where such inspections or tests are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Tests. Inspection and testing of the dispenser/ejector shall be classified as follows:

a. Preproduction Tests

b. Acceptance Tests

4.3 Preproduction Testing.

4.3.1 Testing. Preproduction testing of dispenser(s)/ejector(s) and component parts shall be performed by an independent laboratory acceptable to the procuring activity. The Government reserves the right to witness, modify or

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waiver any of the test requirements specified herein when deemed necessary by circumstances.

4.3.2 Preproduction and Sample Test Reports. Preproduction and sample data shall be prepared and submitted for evaluation in accordance with the requirements of the procuring activity.

4.3.3 Preproduction Test Requirements.

4.3.3.1 Dispenser Assembly. Preproduction tests for both preproduction dispenser samples shall consist of all tests specified herein under "4.7 Test Methods" as listed in Table I with the following exceptions:

a. One ejector unit from each preproduction dispenser shall be subjected to the ejection test requirements of paragraphs 4.7.5.1 and 4.7.6.

b. One ejector unit from each preproduction dispenser shall be subjected to one each locked shut firing test as required by paragraph 4.7.4.

c. One preproduction dispenser shall be subjected to the rocket firing tests of paragraph 4.7.5.2.

d. Each environmental test requirement of Table I shall be performed on only one preproduction dispenser. The test laboratory or the contractor may divide the environmental tests between the two preproduction dispensers to conserve testing time and accommodate scheduling.

4.3.3.2 Ejector Unit Assembly. (When procured separately from the dispenser). Preproduction tests for the ejector unit shall consist of the following requirements:

a. All four preproduction samples shall be subjected to the applicable Individual Test requirements of Table I.

b. Two of the preproduction sample ejectors shall be subjected to the Sampling Test requirements of Table I.

c. One sample ejector shall be subjected to one each locked shut firing test as required by paragraph 4.7.4.

4.4 Acceptance Tests. Acceptance testing shall consist of the following:

a. Individual Tests

b. Sampling Tests

4.4.1 Individual Tests. Each dispenser/ejector or component part shall be subjected to the applicable Individual Tests as listed in Table I.

4.4.2 Sampling Plan and Tests. The sampling plan and sampling tests are applicable to the bomb ejector units only, and shall apply when procured with, or separate from, the dispenser.

4.4.2.1 Sampling Plan. Two sample ejector unit assemblies shall be selected at random by the Government quality assurance representative from each contract lot when lot is less than 150 units, or from each lot of 150 production ejector unit assemblies when larger quantities are involved. Prior to being

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subjected to the sampling tests, the sample ejector units shall have passed the applicable Individual Tests of Table I.

4.4.2.2 Sampling Tests. Sampling tests shall consist of the sampling tests as required by Table I.

4.5 Rejection and Retest Conditions. Failure of the dispenser, bomb ejector, or any component part to satisfactorily complete any applicable preproduction, individual or sampling test requirement as a result of nonconformance to engineering drawing or specification requirements shall be cause for rejection. No dispenser, bomb ejector, or component part on hand or later produced shall be accepted until the extent and cause of failure are determined. The failure shall be corrected and the unit shall demonstrate compliance by satisfactorily passing the failed tests before acceptance of on hand or later produced units. Failure of the dispenser, bomb ejector, or any component part to complete any test requirement as a result of a proven design deficiency shall impose no penalty or restriction upon the manufacturer beyond immediate reporting of the deficiency to the procuring agency.

4.5.1 Defects in Items Already Accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and recommended methods of correcting them.

4.6 Test Conditions. Test conditions shall be as specified in each specific test requirement under "4.7 Test Methods." Unless otherwise specified herein, the dispenser shall be tested with 28 Volt DC. The dispenser shall operate on a minimum of 18 Volt DC.

4.7 Test Methods.

4.7.1 Examination of Product. Prior to packaging, each dispenser/ejector assembly shall be examined visually to determine compliance with the workmanship requirements of this specification. Failure to comply with all requirements shall be cause for rejection.

4.7.2 Conformance to Drawings. Each dispenser/ejector and component part shall be inspected to determine conformance to the engineering drawings, specifications and standards which are applicable. Failure to conform to these criteria shall be cause for rejection.

4.7.3 Functional Test.

4.7.3.1 Ejector Unit Assembly. The ejector units may be inspected on or apart from the dispenser. Manually lock and unlock the ejector unit several times. There shall be no binding between the retainer arms and the housing. There shall be no binding between the piston and its component assembly parts or the housing during any part of the piston travel. The rollers on the retainer arms shall be free to rotate. In the unlocked position, the retainer arms shall remain in position until manually locked. With the retainer arms in the locked position, install safety pin in ejector housing. The pin shall slide freely with no binding, and the arms cannot be manually unlocked. The bomb contact pad shall be locked and not free to rotate on the piston. In the

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locked and unlocked position, the bomb retaining arm springs shall maintain a positive force against the arms such that there is no looseness or free play in either position. The cartridge retainer shall screw easily and smoothly, with no binding, into the housing until fully seated in the cartridge chamber. The firing pin shall protrude .025-.075 inch above the grounding ring and shall always return to its original position after 10 depressions with a suitable tool such as a brass or aluminum rod. The bomb sway brace assemblies shall be moved to their extremes of travel by hand rotating the adjusting nut. Sway braces shall move freely with no binding.

4.7.3.2 Rocket Tube Detent Retention. The rocket tube assembly may be tested for rocket retention on or separate from the dispenser. A MK-43, Mod 1 inert rocket or a simulated rocket, fabricated for this purpose, shall be installed in each tube so as to engage the detent and be subjected to a pull test. The rocket shall not be dislodged from the detent under a forward pull force from 0 to 170 pounds. The rocket shall release from the detent when subjected to a pull force of 170 to 250 pounds.

4.7.4 Locked Shut Firing Test. With the bomb ejector unit assembly mounted on a suitable test stand, install a BDU-33 or MK-106 practice bomb and insert safety pin through the housing. Install an ARD-863-1 cartridge and torque retainer to 250-350 inch pounds. Connect firing pin contact to a 18-30 VDC, 3 Amp (minimum) power supply and a ground lead to the ejector housing. Fire the cartridge. The bomb shall not be ejected and there shall be no damage to the ejector unit. Any evidence of failure or damage shall be cause for rejection. Damage to the safety pin is acceptable. A new safety pin shall be inserted into the locked ejector unit and there shall be no interference or binding.

4.7.5 Reliability Life Test.

4.7.5.1 Bomb Ejector Unit Assembly Ejection Tests. The bomb ejector unit assembly shall be mounted on a suitable test stand capable of withstanding a 1500 pound ejection or reaction force. With the retaining arms adjusted to the 3.87 inch diameter position, 125 releases shall be made using a MK-106 practice bomb. With the retaining arms adjusted to the 4.00 inch diameter position, 125 releases shall be made using a BDU-33 practice bomb. ARD-863-1 cartridges shall be used for all firings. Piston assembly and ejector bore shall be cleaned after each 10 firings or at the end of each firing day. Completion of this test without such damage or erosion which precludes satisfaction of the functional test requirements of paragraph 4.7.3 shall be considered as meeting the operational life requirements of this specification for the ejector unit. Failure to complete this test shall be cause for rejection.

4.7.5.2 Rocket Firing Test. The dispenser shall be mounted from the 14 inch spaced lugs on a suitable test stand securely anchored to withstand blast and thrust effects of the MK-40 or equivalent rocket motor. Test firings shall be conducted at an approved firing range and approved safety procedures shall be observed. There shall be a total of 44 rocket firings and the sequence and mode of the firings shall be as follows:

a. The dispenser shall be loaded with 4 rockets and fired in the ripple mode.

b. The next 36 rockets shall be fired in the singles mode from the number 1 tube.

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TABLE I

TEST	REQUIREMENT PARAGRAPH	TEST PARA-GRAPH	PREPRO-DUCTION TEST	INDI-VIDUAL TEST	SAMPLING TEST
Examination of Product	3.4,3.4.3,3.8	4.7.1	X	X	
Conformance to Drawings	3.4,3.4.3	4.7.2	X	X	
Functional Test	3.5,3.5.5	4.7.3	X	X	
*Locked Shut Firing Test	3.5.4	4.7.4	X		
Reliability Life Test	3.5	4.7.5			
Bomb Ejection Test	3.5	4.7.5.1	X		X
Rocket Firing Test	3.5	4.7.5.2	X		
*Ejection Velocity Test	3.5.2.1	4.7.6	X		X
Environmental Tests	3.6	4.7.7			
High Temperature	3.6.a	4.7.7.1	X		
Low Temperature	3.6.b	4.7.7.2	X		
Temperature Shock	3.6.c	4.7.7.3	X		
Humidity Test	3.6.d	4.7.7.4	X		
Icing Conditions Test	3.6.e	4.7.7.5	X		
Vibrations Test	3.6.f	4.7.7.6	X		
Shock Test	3.6.g	4.7.7.7	X		
Electrical Tests	3.3,3.5.5	4.7.7.8			
Continuity Test	3.3,3.5.5	4.7.7.8.1	X	X	
Dielectric Test	3.3,3.5.5	4.7.7.8.2	X	X	

*These tests may be performed concurrently with the Reliability Life Test in preproduction or sample testing.

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c. The dispenser shall be loaded with 4 rockets and fired in the ripple mode.

The protruding portion of the rocket contact shall be cleaned and manually depressed after each firing to assure proper operation. Replacement of contact is permissible if erosion progresses to the point which prohibits depression or proper contact with rocket. At the conclusion of the test, the forward and aft areas of the end caps, tube and tube attachment points of number 1 tube shall be inspected. Failure to complete this test or observed erosion or structural damage which would render the dispenser unfit or unsafe for further service shall be considered as cause for rejection. Paint or surface finish damage is normal and will not be considered during inspection.

4.7.6 Ejection Velocity Test. Bomb ejection velocity shall be measured and recorded during the bomb ejection testing of paragraph 4.7.5.1. The ejection velocity shall be measured at the end of the piston stroke and shall be an average minimum of 15 feet per second for the 25 pound BDU-33 bomb and an average maximum of 33 feet per second for the 5 pound MK-106 bomb. For each ejector subjected to this test, there shall be a total of 20 velocity measurements made; 10 for the BDU-33 and 10 for the MK-106. For each different bomb configuration, 5 measurements shall be taken from the first 5 firings and the remaining 5 measurements shall be taken from the last 5 firings. Failure of the ejector gun to meet the required velocities for each bomb shall be cause for rejection. Figure 1 is a typical instrumentation set-up for measuring and recording acceleration and velocity and Figure 2 is a typical oscillograph record and sample calculations of the ejector performance characteristics.

4.7.7 Environmental Tests. During or after, as specified, each environmental test the dispenser shall be subjected to an electrical circuit test in accordance with paragraph 4.7.7.8.1 of this specification. The sequence of the different environmental tests is left to the option of the test laboratory. Failure of the dispenser to complete any of the environmental tests shall be governed by the requirements of paragraph 4.5.

4.7.7.1 High Temperature. The dispenser shall be raised to and held at 71°C (160°F) for 8 hours at ambient humidity. The electrical test shall be performed and the temperature reduced to ambient. A visual inspection shall be made for any damage which would cause the dispenser to be inoperable. Evidence of such damage shall constitute a failure.

4.7.7.2 Low Temperature. The dispenser shall be lowered to and held at -53.9°C (-65°F) for 8 hours at ambient humidity. After stabilizing at this temperature, the dispenser shall be subjected to the electrical test. After the temperature has been returned to ambient, a visual inspection shall be made for any damage which would cause the dispenser to be inoperable. Evidence of such damage shall constitute a failure.

4.7.7.3 Temperature Shock. The dispenser shall be tested to the requirements of Method 503, Procedure I, steps 1 thru 5 of MIL-STD-810. At the conclusion of this test the electrical test shall be performed. A visual inspection shall be made for any cracking or rupture of material. Evidence of such damage shall constitute a failure.

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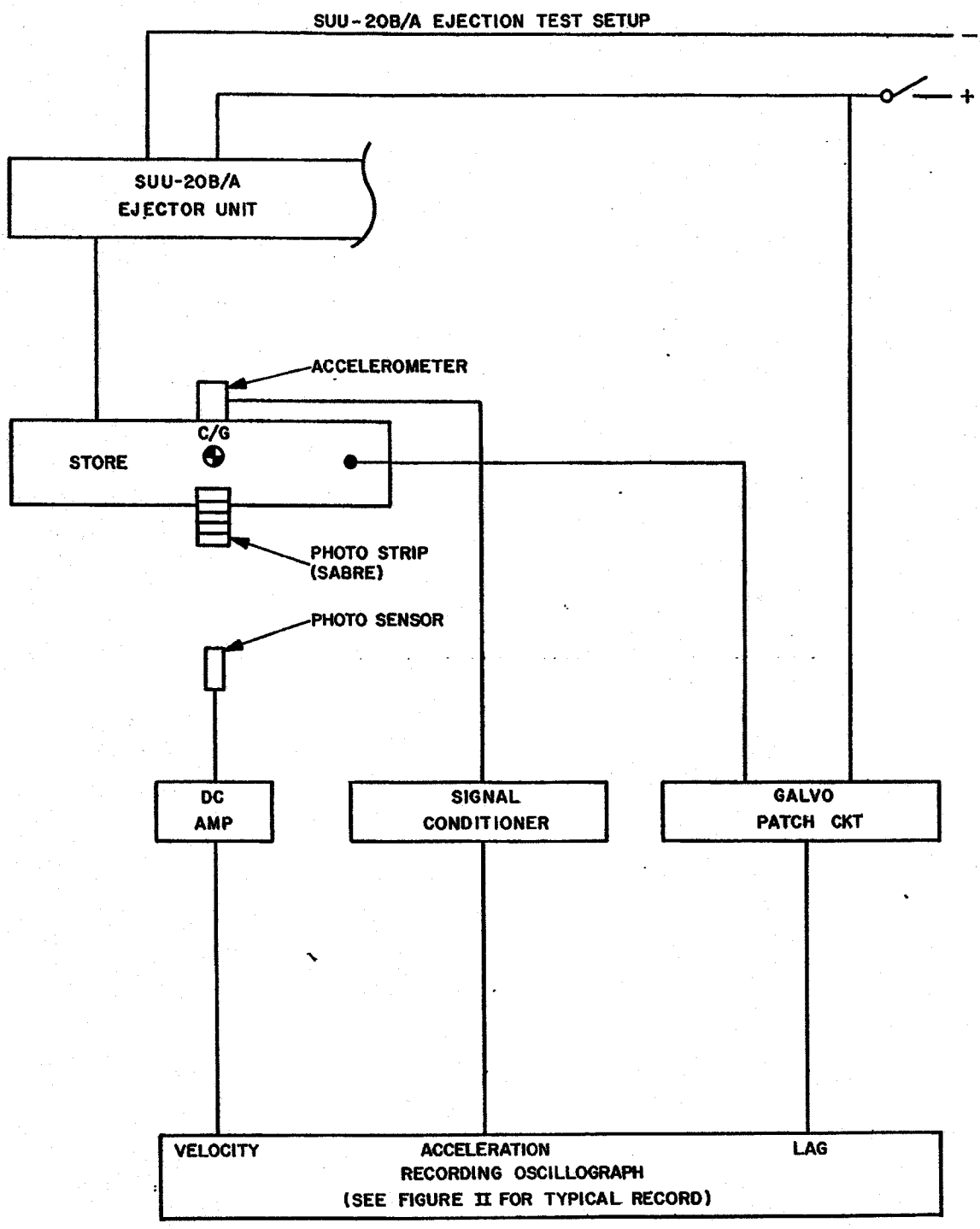


FIGURE I – INSTRUMENTATION SETUP

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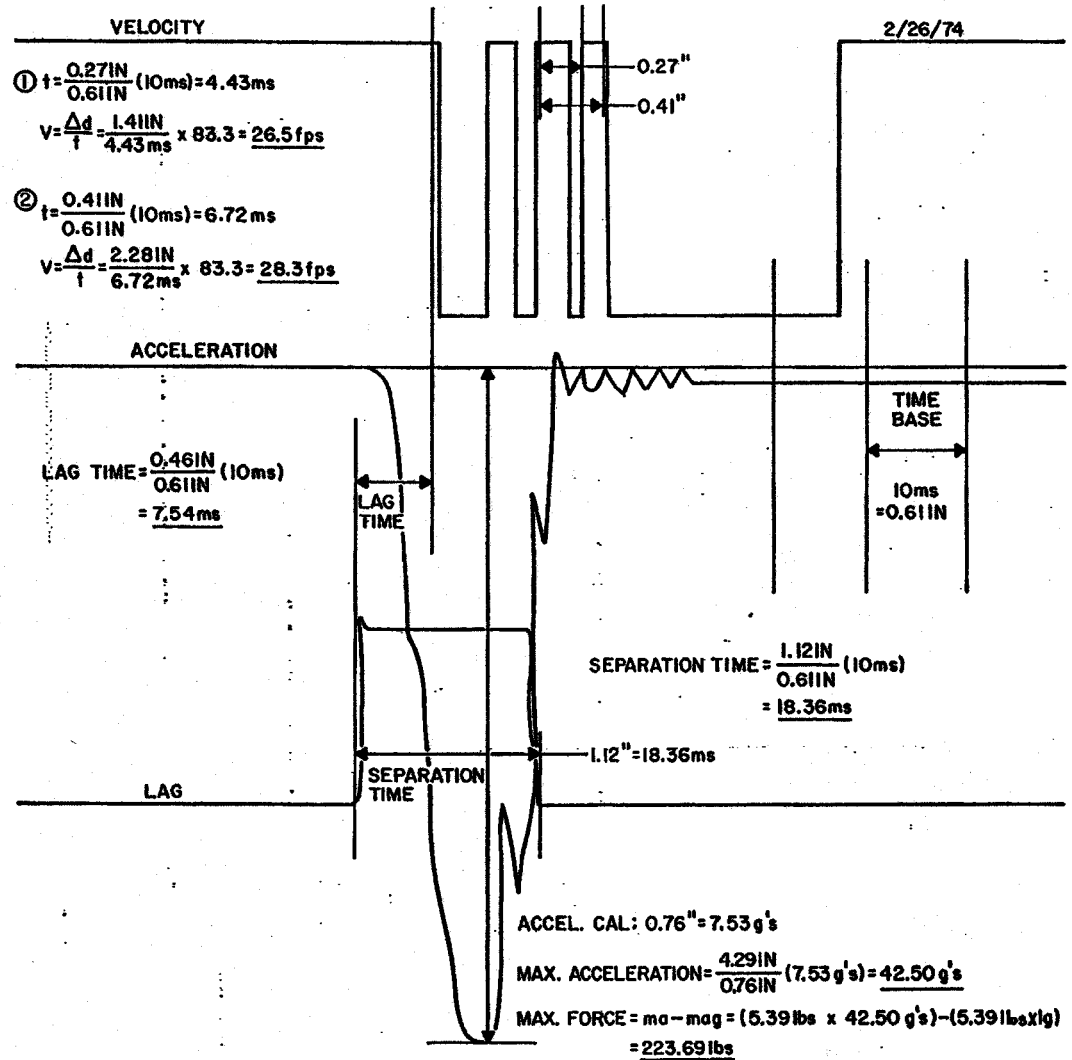


FIGURE 2 SUU-20B/A TYPICAL RECORD & SAMPLE CALCULATIONS (LIGHT STORE)

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4.7.7.4 Humidity. The dispenser shall be tested to the requirements of Method 507, Procedure I, steps 1 thru 6 of MIL-STD-810, except step 5 shall be for 5 cycles (120 hours). At the conclusion of this test the electrical test shall be performed.

4.7.7.5 Icing Conditions. The dispenser shall be stabilized at -53.9°C (-65°F), then raised to 37.7°C (100°F) and 90 per cent relative humidity until frost has disappeared. The dispenser shall again be stabilized at -53.9°C (-65°F). While at this temperature, the electrical test shall be performed.

4.7.7.6 Vibration. The dispenser shall be subjected to sinusoidal and random vibration testing in accordance with the following procedures. Testing shall be performed in each of the three mutually perpendicular axes with the dispenser in the loaded and unloaded configurations. The dispenser shall be suspended from the 14 inch shackles of a MAU-12 bomb ejector rack. The fixture which supports the MAU-12 rack shall be designed to minimize the introduction of undesirable vibration responses into the test set-up. Inert BDU-33 bombs and MK-43 2.75 rocket motors with MK-61 warheads shall be used for loaded testing. Substitution of stores is permissible, providing weights are with ± 3 pounds of the stores specified. Spent ARD-863 cartridges shall be inserted into the retainers and the retainers torqued to 300 ± 50 inch pounds. Torque the BDU-33 bombs only until independent movement is restricted. Excess torque will deform the retaining arms. The MAU-12 sway braces shall be torqued to 450 inch pounds. At the conclusion of each axis testing, the dispenser shall be subjected to the electrical test. Accelerometer locations shall be in accordance with Figure 3. The transverse motion at the input monitoring points shall be minimized, and should be limited to 100 per cent of the input motion. All control accelerometers shall be oriented with their sensitive axis parallel to the direction of the applied vibration. At the conclusion of each vibration mode, the dispenser shall be visually inspected for structural damage which would cause it to be inoperable or unsafe for use. Evidence of such damage shall constitute a failure. Inadvertent release of a bomb during the test shall be considered as a failure.

4.7.7.6.1 Sinusoidal Vibration Testing. The vibration acceleration levels and displacement double amplitudes of the specified curve shall be measured as the averaged signal from the four control accelerometer locations. The sinusoidal vibration testing shall consist of a resonance search only.

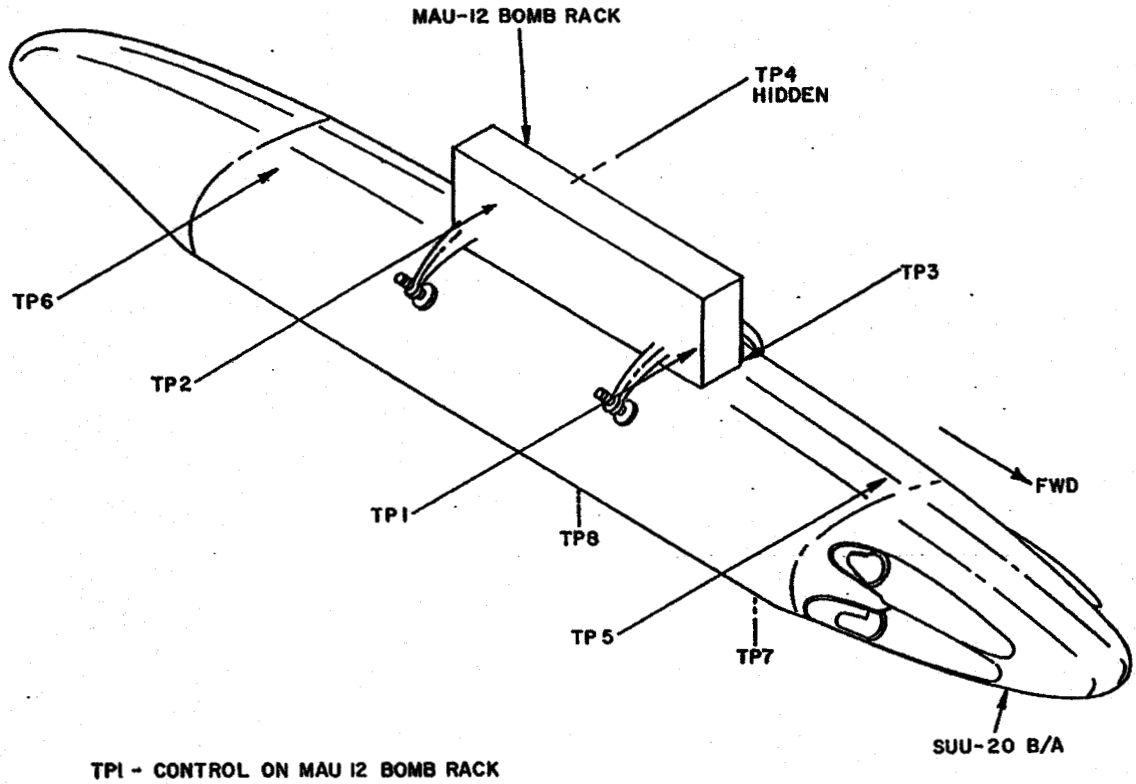
4.7.7.6.1.1 Test Conditions.

a. Unloaded - The dispenser, unloaded and ejectors in the released condition, shall be vibrated through one sweep from 5 to 2000 Hz. The sweep speed through this frequency range shall be 1 octave per minute. The test levels shall be in accordance with the specified curve of Figure 4. The resonance search shall be recorded.

b. Loaded - The dispenser, with stores loaded, shall be vibrated through one sweep from 5 to 2000 Hz. The sweep speed through this frequency range shall be 1 octave per minute. The test levels shall be in accordance with the specified curve of Figure 4. The resonance search shall be recorded.

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SUU-20 B/A VIBRATION TEST ACCELEROMETER LOCATIONS



TP1 - CONTROL ON MAU 12 BOMB RACK
TP2- CONTROL ON MAU 12 BOMB RACK
TP3- CONTROL ON SUU-20 B/A
TP4- CONTROL ON SUU-20 B/A
TP5- TRIAXIAL RESPONSE
TP6- TRIAXIAL RESPONSE
TP7- RESPONSE AT GUN STATION 1 (SENSES DIRECTION OF APPLIED VIBRATION)
TP8- RESPONSE AT GUN STATION 6 (SENSES DIRECTION OF APPLIED VIBRATION)

FIGURE 3

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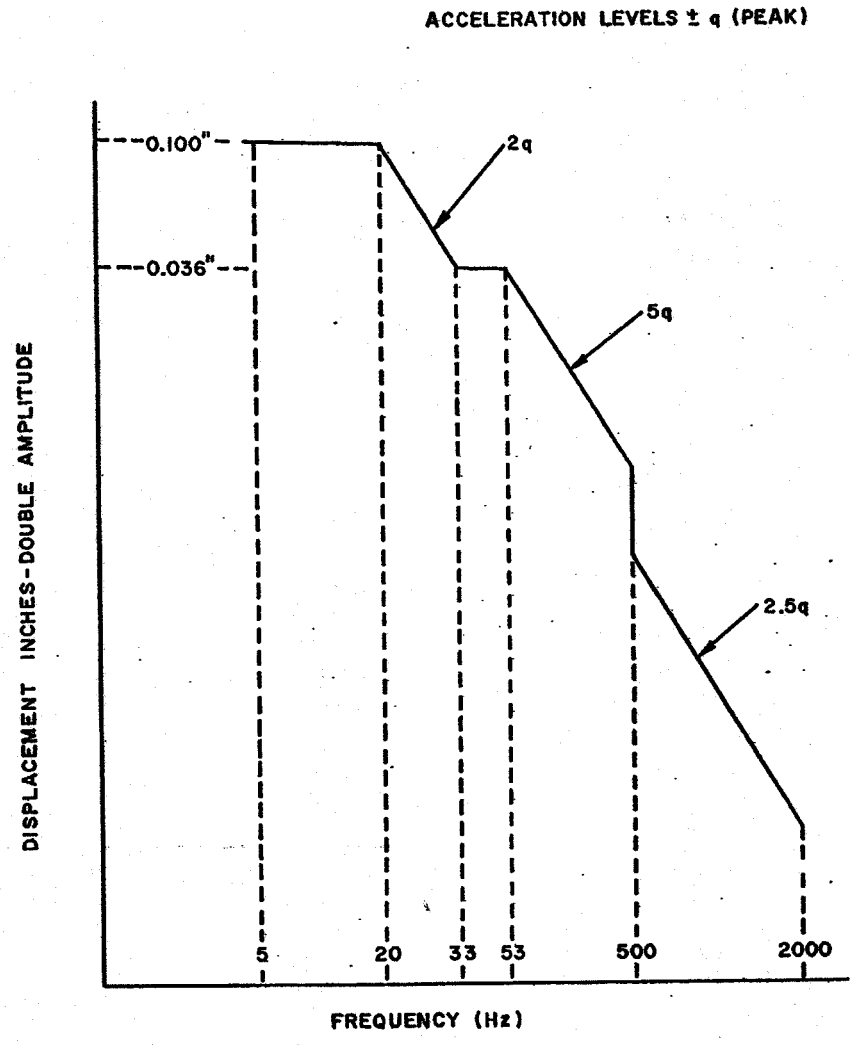


FIGURE 4: SUU-20B/A SINUSOIDAL TEST LEVELS

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4.7.7.6.2 Random Vibration Testing. The ramdom vibration amplitudes shall be measured as the average signal from the four control accelerometer locations. The instantaneous random vibration acceleration peaks shall be limited to three times the RMS acceleration level. The power spectral density analysis of the averaged random vibration control signal shall not deviate from the specified curve by more than +100% or -50% (± 3db) between 20 and 2000 Hz, except that three deviations of +300% (+6db) and three deviations of -75% (-6db), with a total accumulative bandwidth of 100 Hz shall be allowed between 500 and 2000 Hz. Confirmation of these tolerances shall be accomplished by the following analysis parameters with a constant bandwidth analyzer.

20 to 200 Hz - Filter Bandwidth = 20 Hz, Time Constant = 1.5 Seconds, and Sweep Rate = 3.34 Hz per Second.

200 to 2000 Hz - Filter Bandwidth = 50 Hz, Time Constant = 0.6 Seconds, and Sweep Rate = 20.8 Hz per Second.

The composite G - RMS test levels shall not be less than the values specified.

4.7.7.6.2.1 Test Conditions.

a. Unloaded - The dispenser, unloaded and ejectors in the released condition, shall be subjected to a minimum of forty (40) minutes random vibration in accordance with the test levels specified in Figure 5. The test shall be recorded.

b. Loaded - The dispenser, with stores loaded, shall be subjected to a minimum of forty (40) minutes random vibration in accordance with the test levels specified in Figure 5. The test shall be recorded.

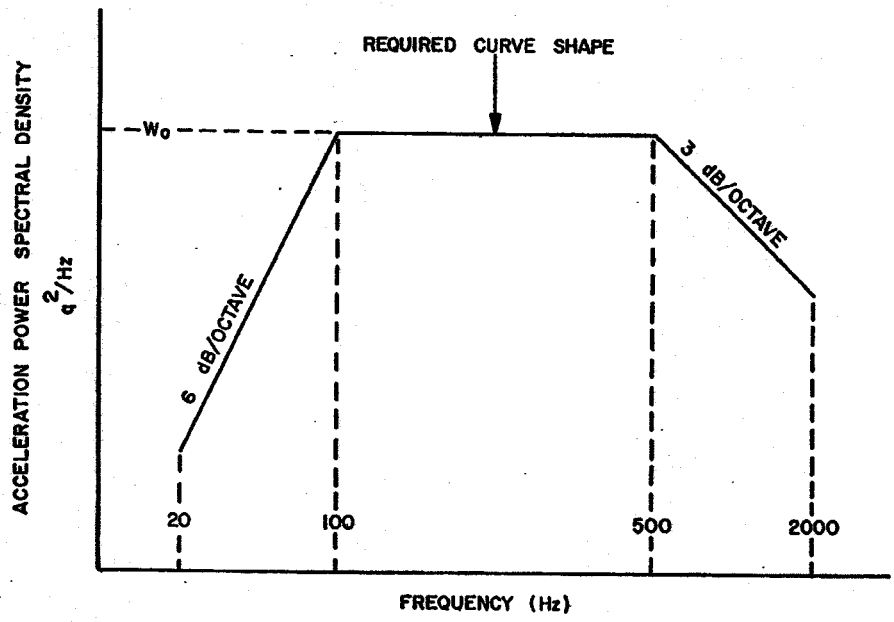
4.7.7.7 Shock Tests. The dispenser shall be mounted from the 14 inch suspension lugs and subjected to an acceleration of 25 ± 5g for 30 ± 10 milliseconds twice in both directions along the vertical, transverse and longitudinal axes. The acceleration wave form shall resemble a half sine wave. The input acceleration shall be monitored on the fixture adjacent to the bomb rack mounting points. The dispenser shall undergo shock tests using the 25 pound BDU-33 practice bombs. No rockets shall be loaded in the dispenser during shock tests. At the conclusion of the shock tests, the electrical test shall be performed.

4.7.7.8 Electrical Tests.

4.7.7.8.1 Continuity Test. The bomb and rocket electrical circuits of the dispenser shall be subjected to a complete continuity test in all select modes. Contractor test equipment and test procedures of Section 8 of T.O. 11B29-3-28-1 may be used for this test.

4.7.7.8.2 Dielectric Test. Pins J and L, K and M, A and M of the internal pod disconnect connector shall be subjected to 1000 volts +50 volts (RMS) 60 CPS for one minute. The dielectric test shall be made with the bomb and rocket circuit disconnect safety spring installed and again with the safety spring removed. Any arc over, breakdown or indication of current flow shall constitute a failure.

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TEST CONDITION	ACCELERATION POWER SPECTRAL DENSITY g^2/Hz W_o	COMPOSITE G-RMS MINIMUM
UNLOADED	0.04	6.6
LOADED	0.02	4.6

FIGURE 5 : SUU-20B/A RANDOM TEST CURVES

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4.8 Quality Control Requirements. The contractor shall provide and maintain a quality control system acceptable to the government for the equipment covered by this specification. The system of quality control shall be in accordance with specification MIL-Q-9858.

4.8.1 Quality Inspection Equipment and Facilities. Inspection equipment and facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspections. The supplier shall establish and maintain a calibration system in accordance with MIL-C-45662.

5. PREPARATION FOR DELIVERY

5.1 The dispenser shall be prepared for delivery in accordance with Transportation, Packaging Order (TPO) 1095-111-4657. The level of preservation, packaging, and packing shall be as specified (See 6.2).

5.1.1 When procured separately, the ejector unit shall be prepared for delivery in accordance with the contract requirements.

6. NOTES

6.1 Intended Use. The dispenser covered by this specification is intended for use on fighter-bomber aircraft for training pilots in the art of bombing and rocketry techniques.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Government furnished property for preproduction and acceptance test purposes (See 3.9).

c. Preproduction test sample (See 3.7).

d. Level of preservation, packaging and packing required (See 5.1).

e. Sampling test sample (See 4.4.2 and 4.4.2.1).

6.3 Gages, Jigs, Fixtures, Dies and Molds. All specific gages, jigs, fixtures, dies or molds peculiar and necessary to the manufacture of the dispenser/ejector assembly shall be provided as required by the contractor and shall become the property of the Government.

6.4 Disposition of Preproduction Dispenser/Ejectors and Sample Test Samples. Preproduction dispensers/ejectors and sample ejector units may be refurbished and shipped to the government as new assemblies providing all component parts and finishes are inspected and found to be in complete compliance with the requirements of this specification and the pertinent drawings and specifications.

6.4.1 Salvaged Pieces. Any piece(s) from a rejected dispenser or ejector unit may be salvaged for use in production providing each salvaged piece is inspected and is found to be in complete compliance with the requirements of this specification and the pertinent drawings and specifications.

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6.5 Cleaning and Lubrication. The dispenser assembly or separate ejector unit assembly shall be cleaned and lubricated in accordance with the requirements of Section 4 of Technical Order 11B29-3-28-1 before shipment. (Technical Order may be obtained from the procuring activity.)

6.6 Symbols. Symbols are not used in this specification revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Custodian:
Air Force-84

Preparing Activity:
Air Force-84

Project No. 1095-F078

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